



*A Project Presentation
on*

Sentinel-X: AI-Based Network Attack Forecasting from Network Traffic Data

Presentation by :

**Shristi Jaiswal — Roll No. 2405250100153
Suraj Prakash Chaudhary — Roll No. 2405250100156
Suraj Upadhyay — Roll No. 2405250100158
Udai Kumar Srivastava — Roll No. 2405250100160**

**Under the guidance of :
Mr. Akash Gupta
(Assistant Professor)**

**Department of Computer Science and Engineering
Buddha Institute of Technology, GIDA, Gorakhpur
Session 2026-27**



Project Content Roadmap



01. Introduction & Shift

Reactive NIDS vs Predictive
Forecasting (SIH26153)

02. Literature & Objectives

Survey of existing NIDS & research
gaps addressed

03. System Pipeline

End-to-end telemetry workflow &
architecture

04. 16 State Features

10s aggregation capturing attack
kinetics

05. Sequence Windowing

Leak-free chronological 70/15/15
splitting

06. LSTM World Model

2-layer backbone with 3 multi-task
heads

07. Autoregressive Rollout

Multi-step rollout ($T+1..T+3$) & joint loss

08. MITRE ATT&CK Lifecycle

5-stage conservative behavioral
progression

09. Explainable AI (SHAP)

Signed risk drivers, inhibitors & temporal
saliency

10. Empirical Benchmarks

100% Recall, 75% FPR reduction &
degradation

11. SOC Command Dashboard

5 interactive views running locally offline

12. Relevance & Conclusion

Critical defense, green AI & future SDN
roadmap



Problem Statement: Shift to Predictive Defense



TRADITIONAL NIDS (REACTIVE)

- **Exploit Execution:**
Attacker launches payload or scan
- **Perimeter Breach:**
System compromised, data exposed
- **Delayed Alert:**
Fires POST-COMPROMISE (Too late)
- **Severe Impact:**
High MTTD (207 days) & alert fatigue

SENTINEL-X (PREDICTIVE FORECASTING)

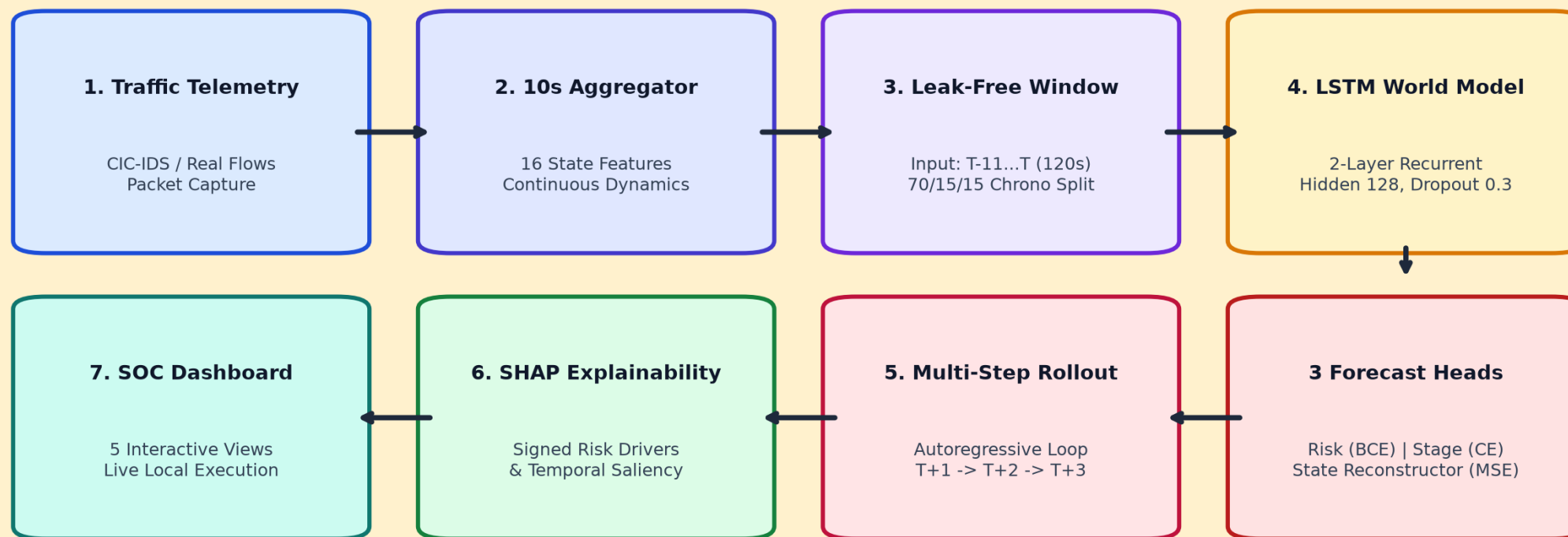
- **Telemetry Tracking:**
Monitors continuous 120s flow window
- **World Model Rollout:**
Forecasts 10s, 20s, 30s ahead
- **Pre-Breach Warning:**
Predicts risk & MITRE stage early
- **Proactive Shielding:**
10-30s buffer to push firewall ACLs

SIH26153 Mandate: Transition from reactive detection (alerting post-breach) to predictive forecasting.

- By predicting risk 10 to 30 seconds into the future, automated firewall policies can isolate malicious hosts before damage occurs.



End-to-End System Pipeline



Closed-Loop Data & Neural Pipeline: Raw Flows → 10s Window Aggregator → 2-Layer LSTM → 3 Heads → Rollout → Dashboard



16-Dimensional Network State Representation



VOLUMETRIC (5 Features)

- flow_count
- total_packets
- packet_rate
- total_bytes
- byte_rate

↳ Detects volumetric floods & exfiltration

TIMING & DURATION (3 Features)

- duration_mean
- duration_std
- iat_mean (inter-arrival time)

↳ Detects rapid scanning & botnet beacon pacing

PACKET DYNAMICS (2 Features)

- fwd_pkt_len_mean
- bwd_pkt_len_mean

↳ Detects asymmetric payloads & auth handshakes

FLAGS & PROTOCOLS (6 Features)

- syn_ratio, rst_ratio
- ack_ratio, fin_ratio
- tcp_ratio, udp_ratio

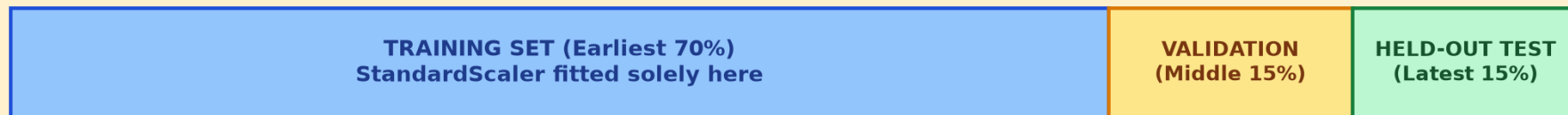
↳ Detects SYN scans, closed ports & flood vectors

Why 10-Second Windows? Packet-by-packet sniffing overflows memory on gigabit links; statistical windowing captures attack kinetics.

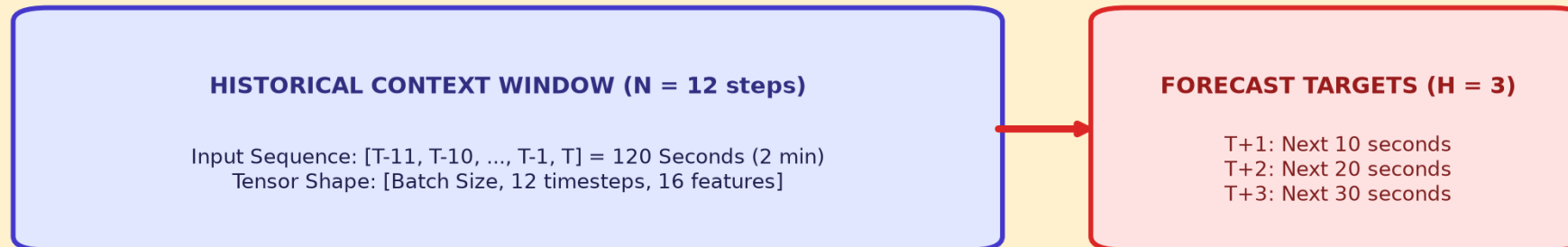


Leak-Free Chronological Windowing

LEAK-FREE CHRONOLOGICAL PARTITIONING (NO RANDOM SHUFFLE)



ROLLING SEQUENCE WINDOW STRUCTURE

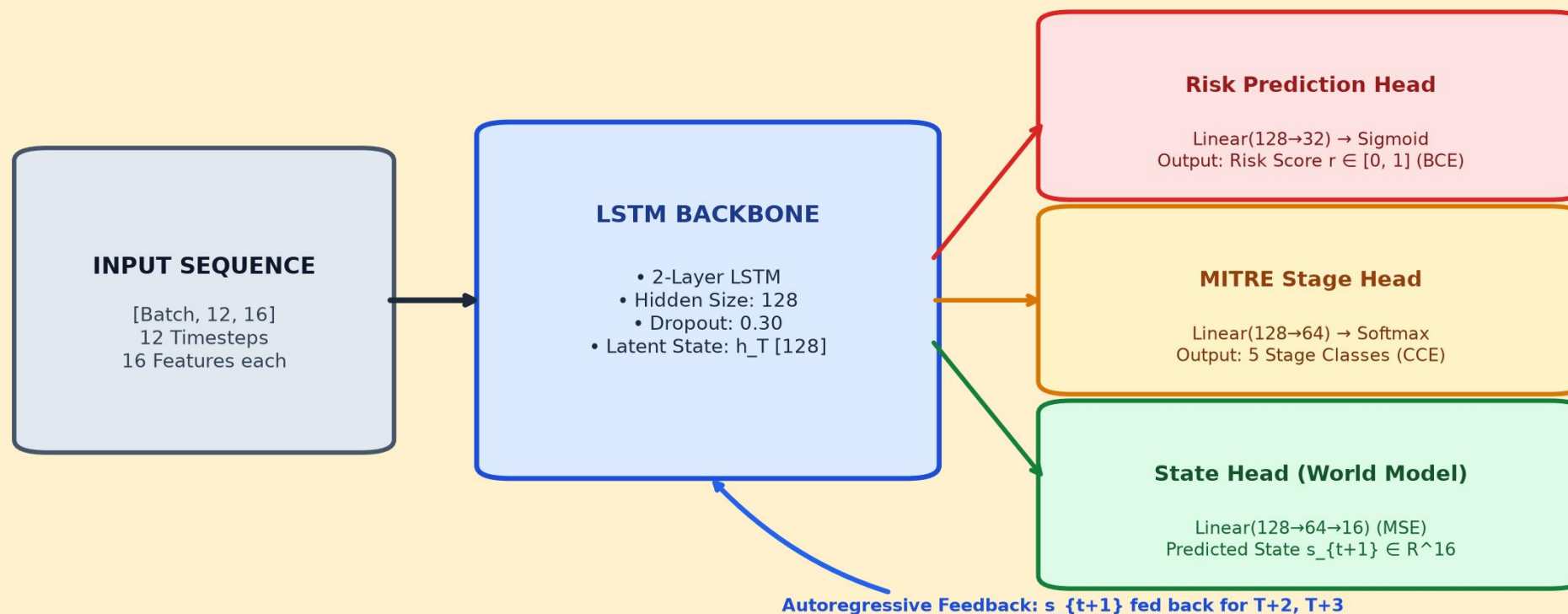


Guaranteed Zero Data Leakage: Standard `train_test_split(shuffle=True)` causes temporal leakage in time series.

- Strict chronological split (Train 70% | Val 15% | Test 15%). Scaler is fitted strictly on the training set.



Deep LSTM World Model Architecture



World Model Paradigm: It doesn't just output a static class; it predicts how the environment itself evolves.

- 2-Layer LSTM (Hidden 128) concurrently drives: Risk Head (BCE), Stage Head (CE), and State Reconstructor (MSE).



Autoregressive Rollout & Multi-Task Loss

AUTOREGRESSIVE MULTI-STEP ROLLOUT

STEP 1: T+1 (Next 10s)

Input: [T-11 ... T]

Output: $s_{\{T+1\}}$, Risk $r_{\{T+1\}}$, Stage $c_{\{T+1\}}$



STEP 2: T+2 (Next 20s)

Input: [T-10 ... T, $s_{\{T+1\}}$]

Output: $s_{\{T+2\}}$, Risk $r_{\{T+2\}}$, Stage $c_{\{T+2\}}$



STEP 3: T+3 (Next 30s)

Input: [T-9 ... $s_{\{T+1\}}$, $s_{\{T+2\}}$]

Output: $s_{\{T+3\}}$, Risk $r_{\{T+3\}}$, Stage $c_{\{T+3\}}$

WEIGHTED MULTI-TASK LOSS

$$\mathcal{L}_{total} = 1.0 \cdot \mathcal{L}_{BCE} + 1.0 \cdot \mathcal{L}_{CE} + 0.2 \cdot \mathcal{L}_{MSE}$$

1. Risk Loss ($w = 1.0$): Binary Cross-Entropy

Penalizes missed attack probability vs benign traffic

2. Stage Loss ($w = 1.0$): Categorical Cross-Entropy

Supervises progression across 5 MITRE behavioral stages

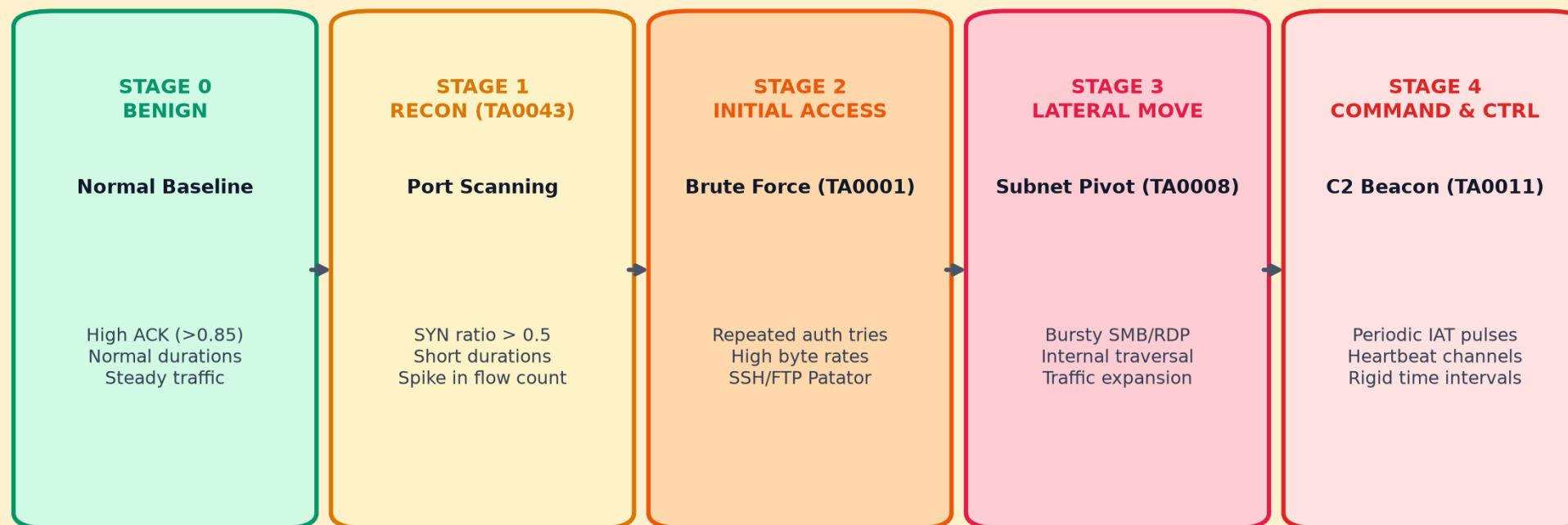
3. Auxiliary State Loss ($w = 0.2$): Mean Squared Error

Forces LSTM to learn real physics of 16 continuous signals

Auxiliary Task Regularization: Predicting continuous state ($w_{state}=0.2$) forces the LSTM to learn real network physics.



Conservative MITRE ATT&CK Lifecycle

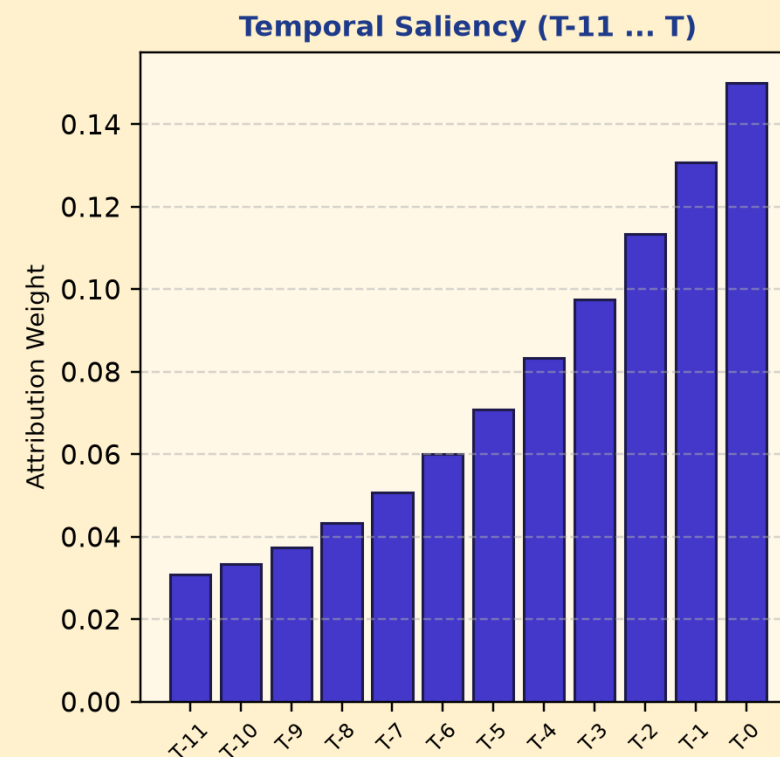
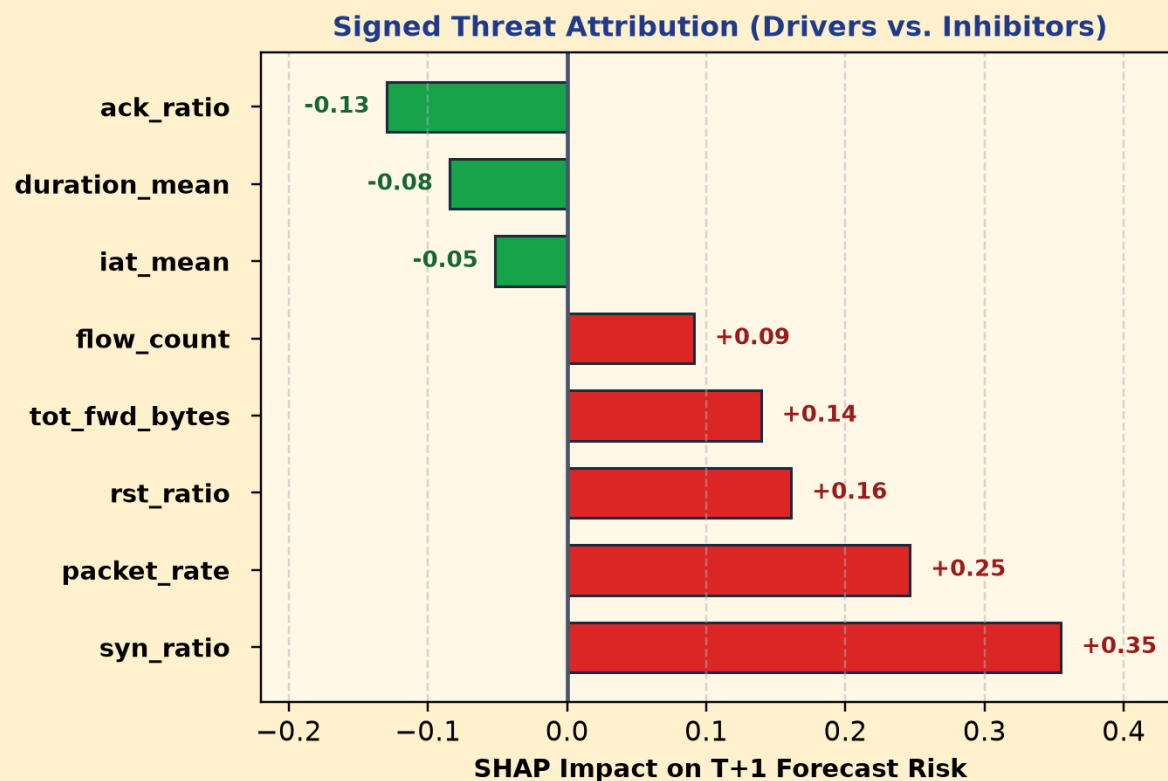


Academic Defensive Posture: We classify behavioral telemetry stages for SOC triage, NOT definitive forensic certainty.

- 5 Distinct Stages: Benign Baseline (0) → Recon (1) → Initial Access (2) → Lateral Movement (3) → Command & Control (4).



Explainable AI (SHAP Threat Attribution)



Eliminating the Black Box: Explains exactly WHY future risk was predicted to escalate.

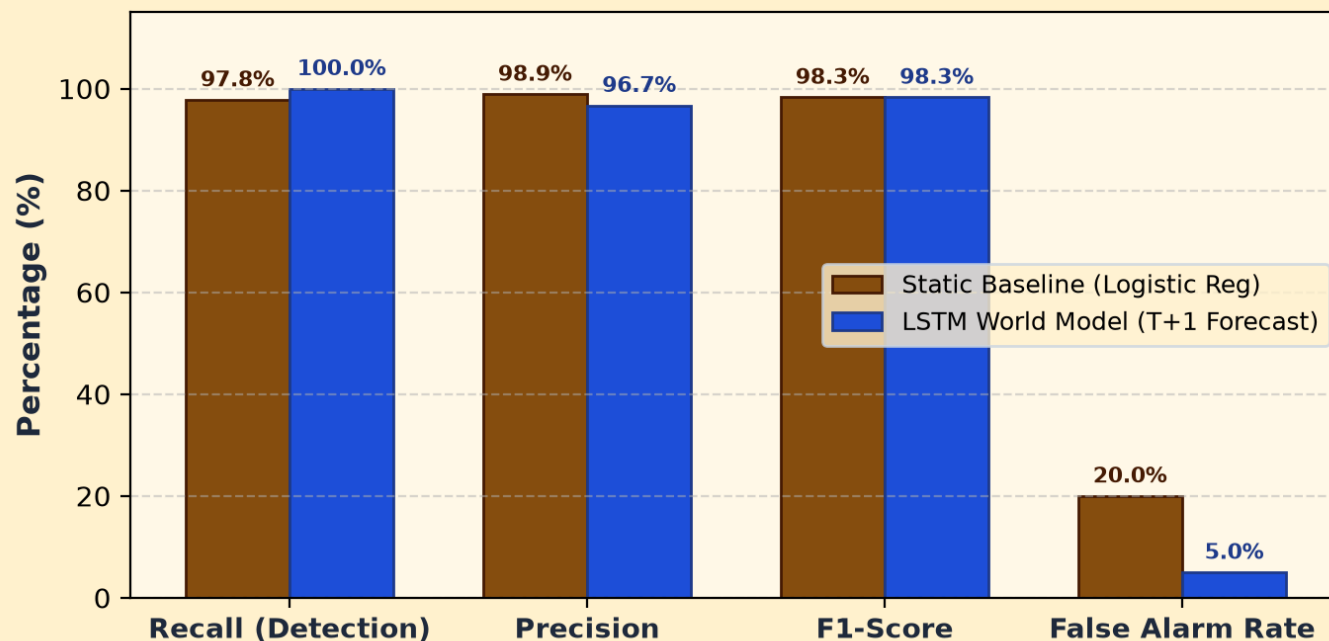
- Red Bars = Risk Drivers (syn_ratio, packet_rate) | Green Bars = Risk Inhibitors (ack_ratio, duration_mean).



Empirical Results: Baseline vs. World Model



Verified Benchmark: Baseline vs. Deep LSTM World Model



100.0% RECALL

Zero missed attacks at T+1
(Baseline missed 2.2%)

75% LESS FALSE ALARMS

FPR reduced from 20% to 5%
Directly solves alert fatigue

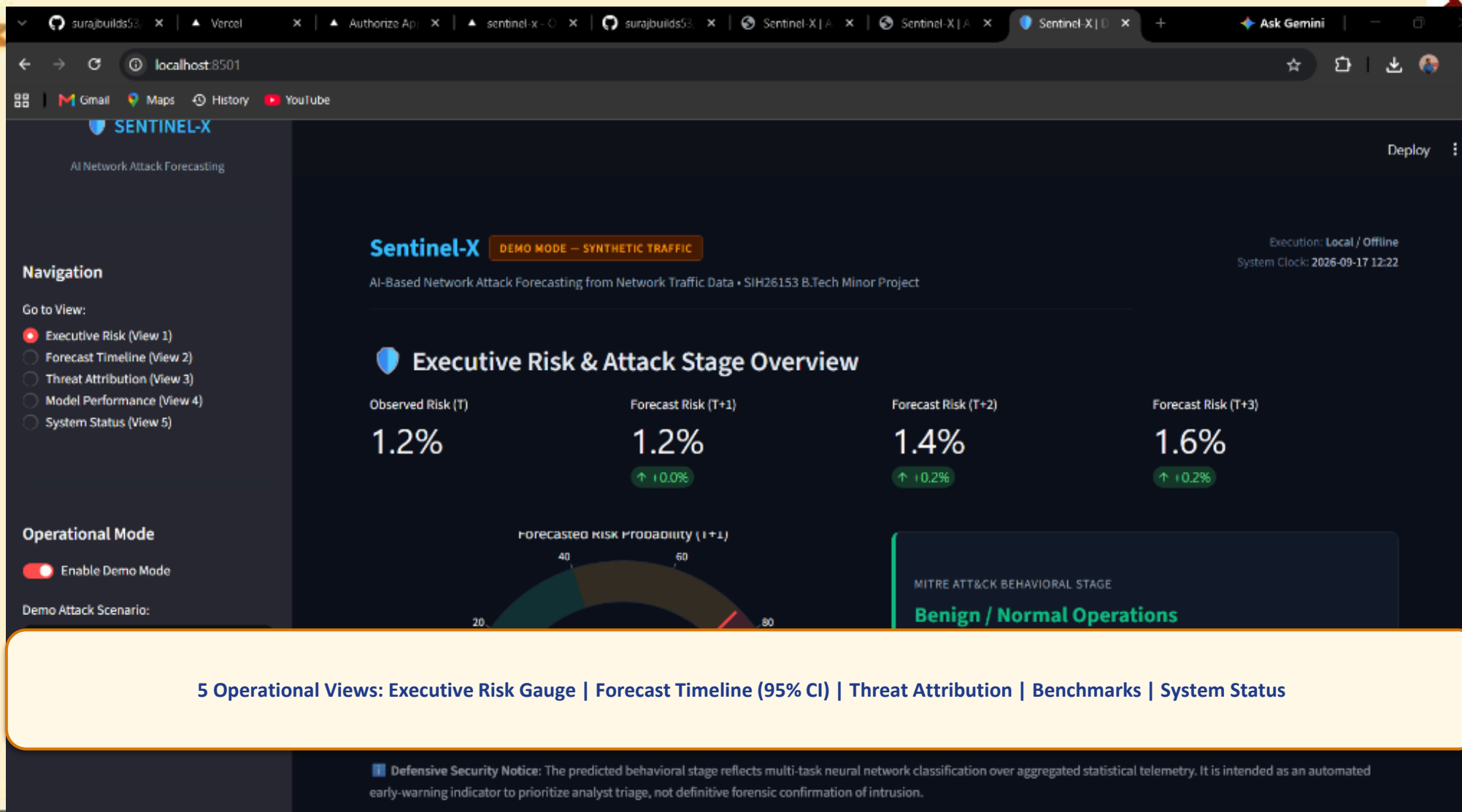
0.9834 F1-SCORE

F1 tapers to 0.9721 at T+3
(Expected horizon decay)

Key Takeaway: Predictive variance compounds naturally as horizon extends, proving genuine autoregressive rollout.



Interactive SOC Command Dashboard





Societal Impact & Future Roadmap



PILLAR 01

AUTOMATED SDN ORCHESTRATION

- ✓ Direct API linkage to OpenFlow / Ryu
- ✓ Pushes automated firewall ACL rules
- ✓ Pre-emptively drops malicious IPs
- ✓ Sub-second autonomous containment

PILLAR 02

TEMPORAL GRAPH NETWORKS (T-GNN)

- ✓ Models host-to-host lateral topology
- ✓ Captures subnet pivot escalation
- ✓ Graph embeddings + LSTM temporal
- ✓ Prevents multi-hop ransomware

PILLAR 03

EDGE ROUTER QUANTIZATION

- ✓ ONNX & TensorRT INT8 quantization
- ✓ Inference latency dropped to < 5 ms
- ✓ Deployable inside edge gateways
- ✓ Zero cloud dependency / green AI

Green AI Architecture: Single-sample inference latency < 20 ms on standard laptop CPU with zero cloud costs.

- Proactive shielding for critical infrastructure (power, banking, healthcare) against multi-phase ransomware & breaches.



THANK YOU

Questions & Discussion

Shristi Jaiswal | Suraj Prakash Chaudhary | Suraj Upadhyay | Udai Kumar Srivastava

Under the guidance of: Mr. Akash Gupta (Assistant Professor)

Department of Computer Science and Engineering

Buddha Institute of Technology, GIDA, Gorakhpur

GitHub: github.com/surajbuilds53/sentinel-x | Live Hub: sentinel-x-lilac.vercel.app